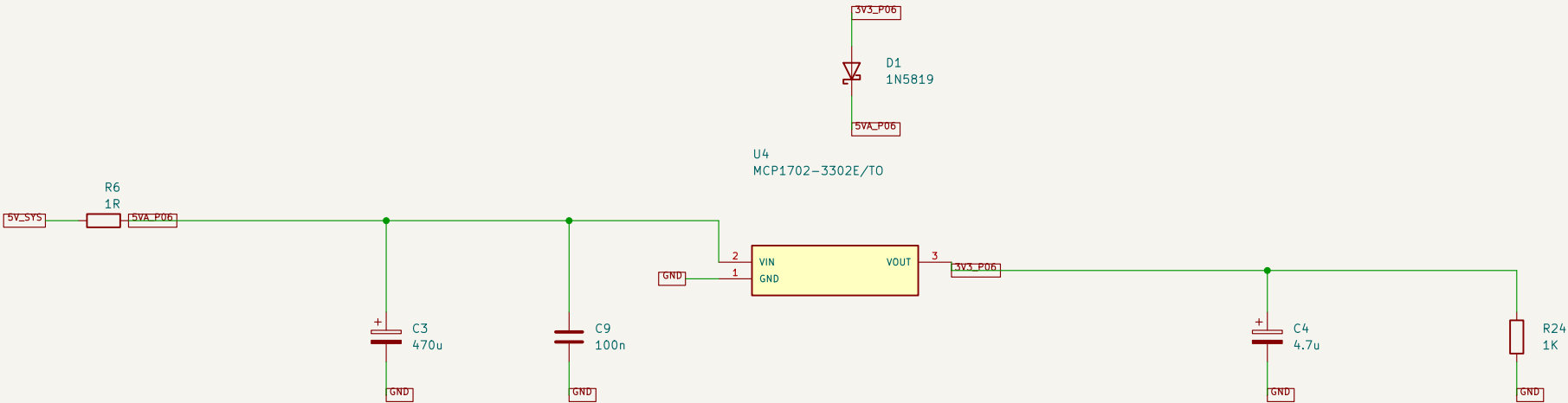
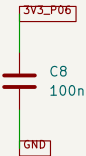
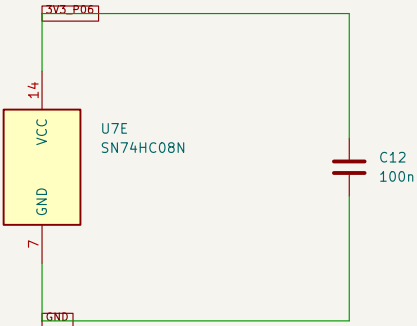
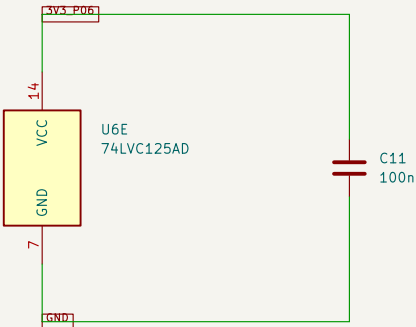
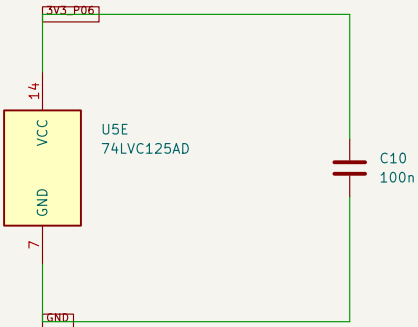
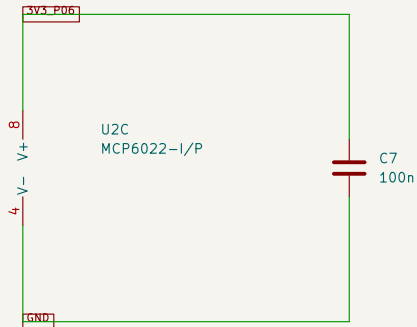




LV06: 5V_SYS zasila plytkę. 3V3_IO z P02 jest tylko na TP4. Logika pracuje z lokalnego 3V3_P06.



FORCE

ANA

DIG

READY

CONNECT

#FLG1
PWR_FLAG

#FLG2
PWR_FLAG

#FLG3
PWR_FLAG

#FLG4
PWR_FLAG

D1: anoda 3V3_P06, katoda 5VA_P06. Budzet MEASURE: 180 mA, wiekszosc poboru przez R21 obciazajacy styk.

Inline current measurement; review, hardware not tested

Prototype / schematic + PCB review

Sheet: /
File: P06.kicad_sch

Title: EGRLab P06 I-LOGGER

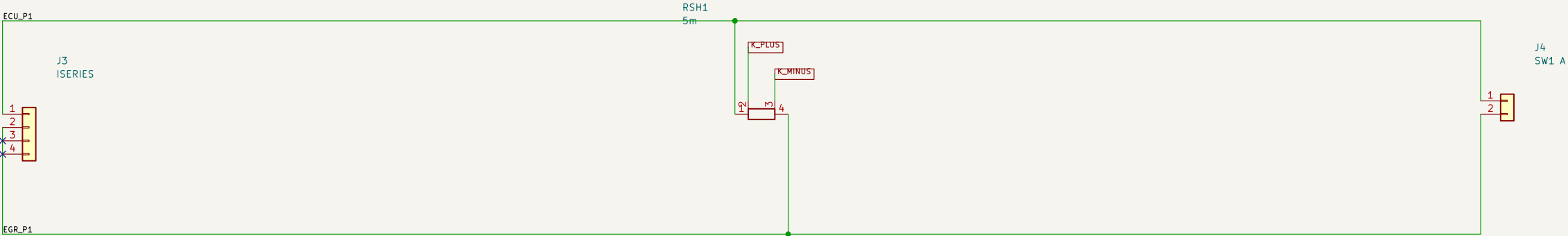
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-27

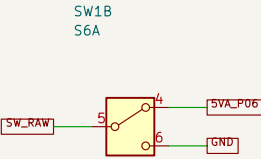
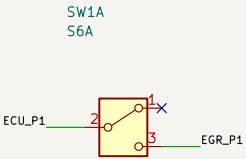
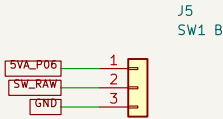
Rev: P06-R1

Id: 1/6

RSH1 stale w torze ECU_P1 -> EGR_P1. Brak przełącznika szeregowego, przekaznika i połączenia mocy z GND.



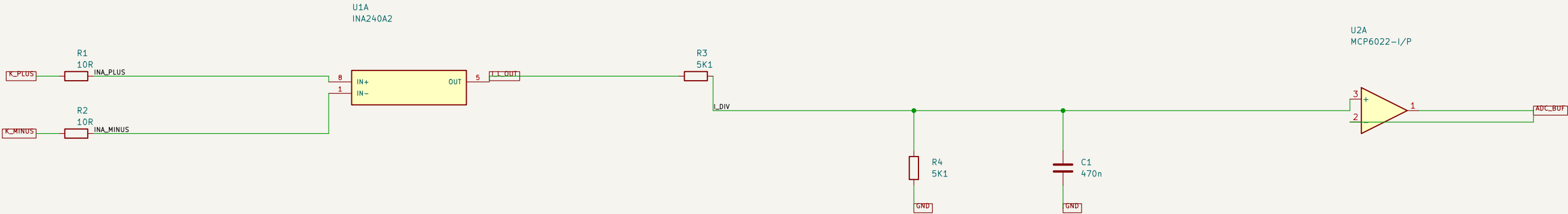
BYPASS: SW1 2-3 + 5-6. MEASURE: SW1 2-1 (NC) + 5-4. COM to zaciski 2 i 5, nie 1 i 4.



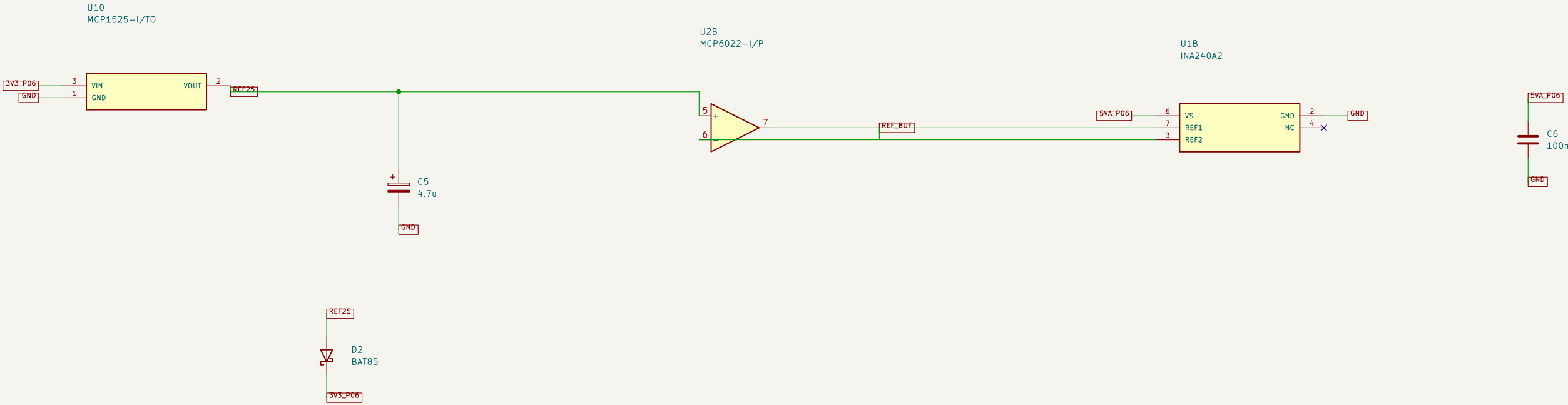
SW1 poza PCB. J4.1 -> SW1.2; J4.2 -> SW1.3. J5.1 -> SW1.4; J5.2 -> SW1.5; J5.3 -> SW1.6.

R21: 39R / 2W, prąd styku około 0.12 A. Gorący element oddalony od bocznika. Nie przełączać przy włączonym zaplonie.

5 mOhm x 50 V/V x 1/2 = 0.125 V/A. ADC: 1.25 V przy 0 A, idealnie 204.8 kodu/A; kalibracja offsetu i nachylenia obowiązkowa.



R3 || R4 = 2.55k; C1 = 470n PET: fc = 133 Hz, tau = 1.20 ms. Pomiar trendu prądu; nie przebieg pojedynczych impulsów PWM.



U1 SOIC D: 8 IN+, 1 IN-, 5 OUT, 6 VS, 2 GND, 7 REF1, 3 REF2, 4 NC. Oba REF sterowane wspólnym buforem.

U10 T092: 1 GND, 2 VOUT, 3 VIN. C5 4.7uF przy U10; R1/R2 10R 0.1% identyczne, szczytki Kelvin lokalne.

Inline current measurement; review, hardware not tested

Prototype / schematic + PCB review

Sheet: /ANA/

File: ANA.kicad_sch

Title: EGRLab P06 I-LOGGER

Size: A3

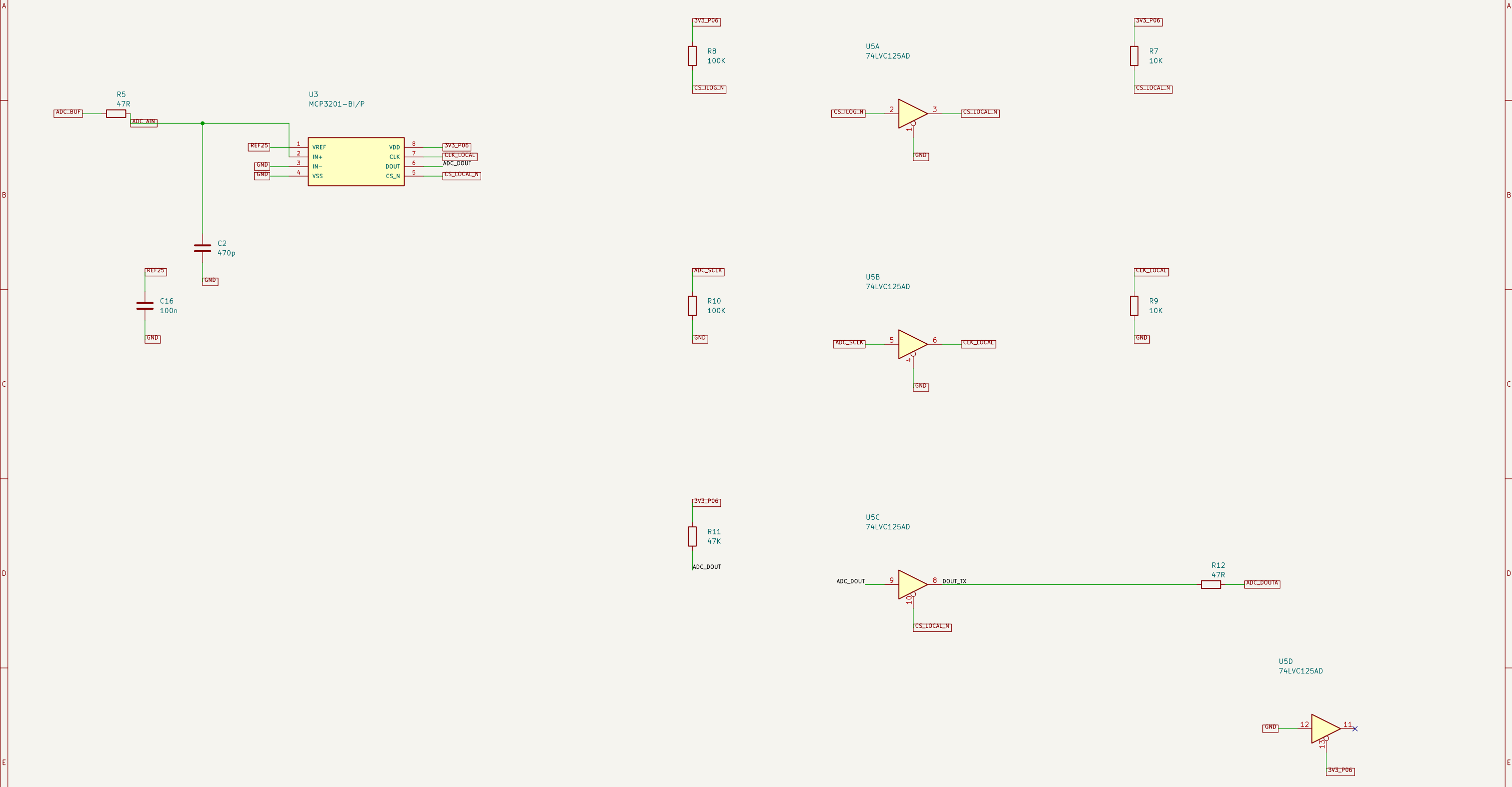
Date: 2026-09-27

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```
SPI mode 0, 500 kHz, 16 zegarów. raw = (word >> 1) & 0xFFFF. DOUT odłączony od wspólnej magistrali gdy CS_LOCAL_N=1.
```

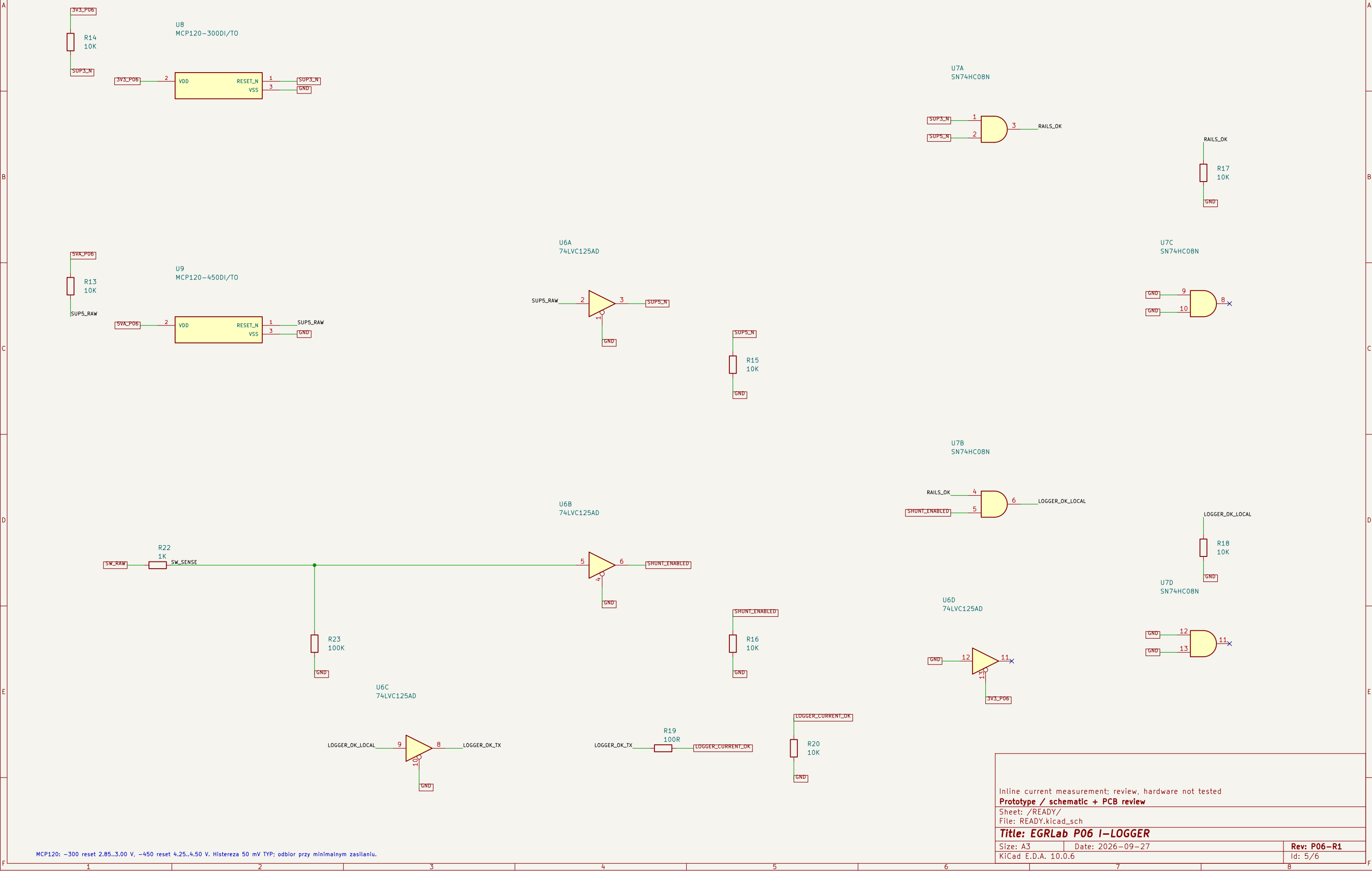


2 kS/s z firmware v6.1. Probka niesymultaniczna wzgledem AD7606B; zachowac timestamp/skew. Nie wyprowadzac I2C do analogu.

U5/U6 tylko Nexperia 74LVC125AD z 1off. HC125 nie jest zamiennikiem. SOIC/SO14 raster 1.27 mm.

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LOGGER_CURRENT_OK = SUP3_N & SUP5_N & SHUNT_ENABLED; BYPASS daje LOW. READY nie jest dowodem kalibracji ani sprawnosci bocznika.



Inline current measurement; review, hardware not tested

Prototype / schematic + PCB review

Sheet: /READY/
File: READY.kicad_sch

Title: EGRLab P06 I-LOGGER

Size: A3	Date: 2026-09-27	Rev: P06-R1
KiCad E.D.A. 10.0.6		Id: 5/6

